SUPPLEMENTARY MATERIALS

A. Modeling Details

In the implementation of FAST on our self-collected dataset, we used $L_t = 4$, $M = 8$ and $F = 32$, for the F_i feature vector, its dimension will be $8 \times 32 = 256$, and $L = 4$ for the transformer layers. The AdamW optimizer was employed with an initial learning rate of 0.001. The learning rate schedule follows a cosine decay pattern with a warm-up phase. During the first 10 epochs, the learning rate gradually increases from 10% to 100%, enabling a smooth warm-up period. Following this, the learning rate decays from 100% to 10% over the remaining training epochs. To accelerate the pre-training phase, FP16 automatic mixed precision training was used, conserving memory and enhancing GPU throughput. For the fine-tuning phase, FP32 precision was utilized without mixed precision. We use two A100 GPUs, the experiment can be completed in approximately 5 hours.

B. Metrics for Evaluation

$$\textbf{Accuracy:} \quad \frac{TP+TN}{TP+TN+FP+FN}$$

$$\textbf{Precision (Macro):} \quad \frac{1}{C}\sum_{i=1}^{C}\frac{TP_i}{TP_i+FP_i}$$

$$\textbf{Recall (Macro):} \quad \frac{1}{C}\sum_{i=1}^{C}\frac{TP_i}{TP_i+FN_i}$$

$$\textbf{F1-score (Macro):} \quad \frac{1}{C}\sum_{i=1}^{C}\frac{2\cdot \text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}$$

$$\textbf{Cohen's Kappa:} \quad \kappa = \frac{p_o - p_e}{1-p_e}$$

$$p_o = \frac{1}{N}\sum_{i=1}^{N}\delta(y_i,\hat{y}_i), \quad p_e = \sum_{k=1}^{C}\left(\frac{n_k}{N}\cdot\frac{m_k}{N}\right)$$

Where p_o is the observed agreement, p_e is the expected agreement, n_k and m_k are the actual and predicted counts for class k, respectively.

$$\textbf{AUC (One-vs-Rest):} \quad \text{AUC} = \frac{1}{C}\sum_{i=1}^{C}\text{AUC}_i$$

Where AUC_i is the area under the ROC curve for class i, computed using one-vs-rest strategy.

Explanation of Symbols:

- C: Number of classes in the classification task.
- $\delta(y_i, \hat{y}_i)$: Indicator function that is 1 if $y_i = \hat{y}_i$ (correct prediction) and 0 otherwise.
- N: Total number of samples in the dataset.
- n_k: Actual number of samples belonging to class k.
- m_k: Predicted number of samples assigned to class k.
- p_o: Observed agreement - the proportion of times the predicted label matches the true label.
- p_e: Expected agreement - the agreement expected by chance, based on class distributions.
- AUC_i: The area under the ROC curve for class i, computed using the one-vs-rest strategy.

C. Visualization Details

To interpret model predictions, the Integrated Gradients (IG) method [1] is employed. This method assigns an importance score to each input feature by computing the path integral of the gradients of the model's output with respect to the input, along a straight line path from a pre-defined baseline to the actual input. The IG method satisfies two key axioms: Sensitivity, which ensures that features contributing differently to the output are assigned different importance scores, and Implementation Invariance, which guarantees that two functionally equivalent models yield the same importance scores. The IG computation is formalized as:

$$IG_i(x) = (x_i - x_i')\int_{\alpha=0}^{1}\frac{\partial F(x' + \alpha(x - x'))}{\partial x_i}d\alpha$$

where x is the input, x' is the baseline, F is the model's output function, and i denotes the i-th feature. This method provides an intuitive understanding of the contribution of each input feature to the model's prediction, as illustrated in Figure **??**.

D. Results on Overt Speech Decoding

the decoding accuracy for overt speech is higher compared to covert speech, as seen in Table I. This difference can be attributed to two key factors. First, overt speech likely produces stronger and more consistent neural activity due to the physical act of speaking, which enhances the reliability of the decoding process. Second, the muscle movements involved in overt speech may generate electromyographic signals, providing additional information that aids in classification.

E. Ablation on Brain Areas Partitioning Configurations

In addition to the M=8 in the manuscript, here we add the following configurations: (a) M=5 by merging the electrodes from prefrontal into frontal, precentral, and postcentral into central; (b) M=4 by additionally merging parietal into occipital; (c) M=3 by only keeping the frontal, central and the occipital. The performance is shown in the Table II (a)(b)(c). The performance metrics generally decrease as the number of regions M decreases from FAST to conditions (a)(b) and (c).

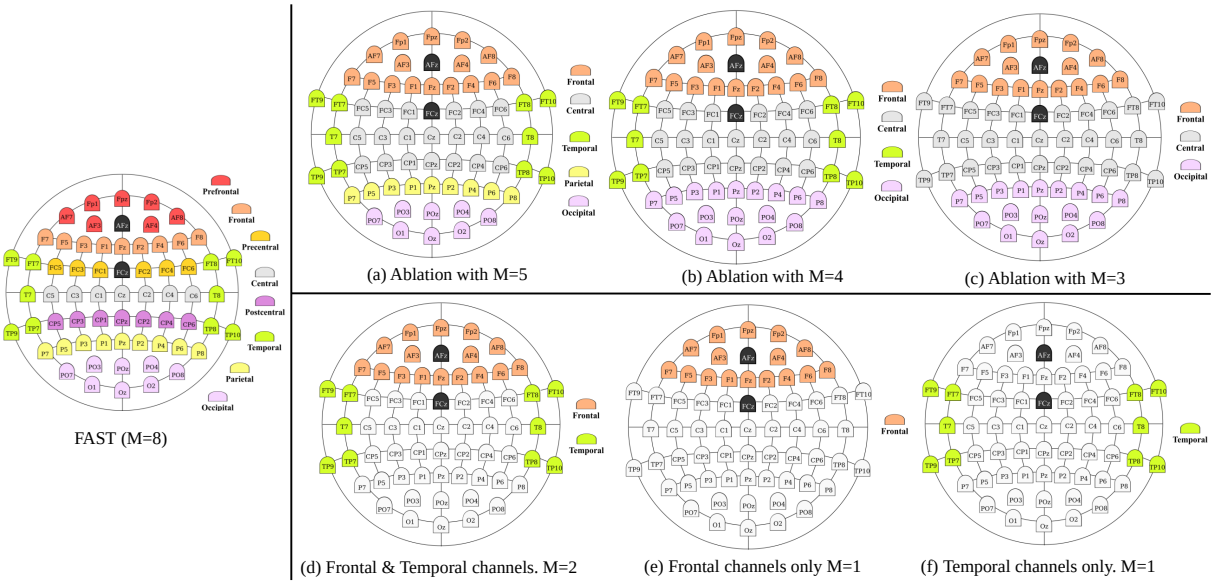
To examine the impact of using only speech-related electrodes, three additional ablation studies were conducted: (d) using both Frontal and Temporal electrodes, (e) using Frontal electrodes only, and (f) using Temporal electrodes only. As shown in Table II (d)(e)(f), the baseline FAST model (M=8) outperforms all ablated configurations in both pre-train and finetune stages. Among the ablations, temporal electrodes were shown more impactful during finetuning as shown in (c).

F. Additional Public Dataset

In addition to the self-collected dataset, to validate the effectiveness of FAST, we included the publicly available multi-class imagined speech classification dataset from the 2020 International BCI Competition [2], this dataset consists of EEG recordings from 15 subjects, aged 20-30 years old. The subjects were instructed to imagine the silent pronunciation of five conversational words/phrases: 'hello,' 'help me,' 'stop,'

TABLE I
PERFORMANCE COMPARISONS COVERT/OVERT SPEECH DECODING FOR 5 UTTERANCES

Stage	Condition	Accuracy	F1-score	Cohen-Kappa	AUC
Pretrain	Covert	26.9 ± 6.3	0.221 ± 0.091	0.087 ± 0.086	0.627 ± 0.083
	Overt	30.1 ± 8.8	0.234 ± 0.113	0.126 ± 0.111	0.681 ± 0.096
Finetune	Covert	34.7 ± 10.7	0.340 ± 0.108	0.184 ± 0.134	0.662 ± 0.097
	Overt	51.3 ± 12.9	0.523 ± 0.104	0.496 ± 0.089	0.721 ± 0.058



FAST (M=8)

(a) Ablation with M=5

(b) Ablation with M=4

(c) Ablation with M=3

(d) Frontal & Temporal channels. M=2

(e) Frontal channels only M=1

(f) Temporal channels only. M=1

Fig. 1. illustration of different M and configurations. (a) M=5 by merging the electrodes from the prefrontal into frontal, precentral, and postcentral central; (b) M=4 by additionally merging the parietal into occipital; (c) M=3 by only keeping frontal, central, and occipital. (d) using both Frontal and Temporal electrodes, (e) using Frontal electrodes only, and (f) using Temporal electrodes only.

TABLE II
PERFORMANCE COMPARISONS WITH DIFFERENT M AND CONFIGURATIONS

Stage	Model	Accuracy	F1-score	Cohen-Kappa	AUC
Pretrain	FAST (M=8)	26.9 ± 6.9	0.221 ± 0.091	0.087 ± 0.086	0.627 ± 0.083
	(a) M=5	25.5 ± 5.2	0.163 ± 0.067	0.069 ± 0.065	0.633 ± 0.073
	(b) M=4	25.6 ± 5.5	0.168 ± 0.065	0.071 ± 0.068	0.634 ± 0.076
	(c) M=3	25.9 ± 5.6	0.178 ± 0.074	0.074 ± 0.070	0.637 ± 0.074
	(a) M=2 Frontal + Temporal	26.9 ± 5.5	0.195 ± 0.072	0.087 ± 0.069	0.638 ± 0.072
	(b) M=1 Frontal Only	26.4 ± 5.1	0.180 ± 0.068	0.080 ± 0.064	0.637 ± 0.071
	(c) M=1 Temporal Only	27.4 ± 6.4	0.215 ± 0.079	0.092 ± 0.080	0.642 ± 0.072
Finetune	FAST (M=8)	34.7 ± 10.7	0.340 ± 0.108	0.184 ± 0.134	0.662 ± 0.097
	(a) M=5	33.1 ± 7.0	0.291 ± 0.073	0.164 ± 0.087	0.645 ± 0.080
	(b) M=4	32.7 ± 7.8	0.288 ± 0.082	0.159 ± 0.097	0.643 ± 0.081
	(c) M=3	33.4 ± 7.2	0.293 ± 0.072	0.167 ± 0.090	0.652 ± 0.077
	(d) M=2 Frontal + Temporal	33.7 ± 8.1	0.299 ± 0.083	0.172 ± 0.102	0.652 ± 0.083
	(e) M=1 Frontal Only	33.8 ± 8.4	0.289 ± 0.087	0.172 ± 0.105	0.649 ± 0.078
	(f) M=1 Temporal Only	34.2 ± 8.7	0.316 ± 0.087	0.178 ± 0.109	0.651 ± 0.078

'thank you,' and 'yes.' given the word as if they were performing real speech, without moving any articulators or making the sound. Each subject participated in 70 trials per class, yielding a total of 350 trials (60 trials per class for training and 10 trials per class for validation). EEG signals were recorded from 64 electrodes following the 10-20 international system. Ground and reference electrodes were placed at Fpz and FCz.

For this dataset, the limited subject count (15 subjects) led to subject-independent pre-training being ineffective for both FAST and baseline models. Therefore, we proceeded directly with subject-dependent training, performing a 5-fold cross-validation on each subject's 350 trials.

G. Results on public dataset

Table III provides the classification accuracies and standard deviations across various models on the BCI competition dataset, the dataset only contains a single utterance per trial. The table also presents the results of the Wilcoxon signed-rank test, with significant p-values marked by asterisks. The individualized accuracy for [6] is unavailable and is indicated as (NA). FAST achieved the highest average accuracy at 54.8% with a standard deviation of 9.1%. Among the models

TABLE III
PERFORMANCE COMPARISONS ON 2020 INTERNATIONAL BCI COMPETITION DATASET, TRACK 3: IMAGINED SPEECH CLASSIFICATION

Method	Accuracy (1 utterance)
LSTM	36.5 ± 4.8 ***
GRU	38.3 ± 4.7 ***
DeepConvNet [3]	33.2 ± 5.8 ***
EEGNet [4]	40.6 ± 6.2 ***
EEG-Conformer [5]	43.3 ± 5.1 ***
Dong-Yeon et al. [6]	48.1 ± 3.6 (NA)
TSception [7]	52.2 ± 8.8 ***
FAST	**54.8 ± 9.1**

Note *** denoting a p-value less than 0.001, ** for less than 0.01, and * for less than 0.05

evaluated, EEG-Conformer, [6], and TSception also demonstrated competitive performance, achieving average accuracies of 43.38%, 48.10%, and 52.26% respectively. The FAST model outperformed all other approaches.

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